Project: Physics-Informed Neural Networks for Battery Degradation Modeling

Introduction

In this project, we explore how neural networks and physics-informed neural networks (PINNs) can be used to model accelerated battery aging.

Lithium-ion batteries often exhibit nonlinear capacity loss, especially during accelerated aging. In this simplified example, we simulate a trajectory where the battery capacity decreases faster over time. This represents an accelerating degradation process.

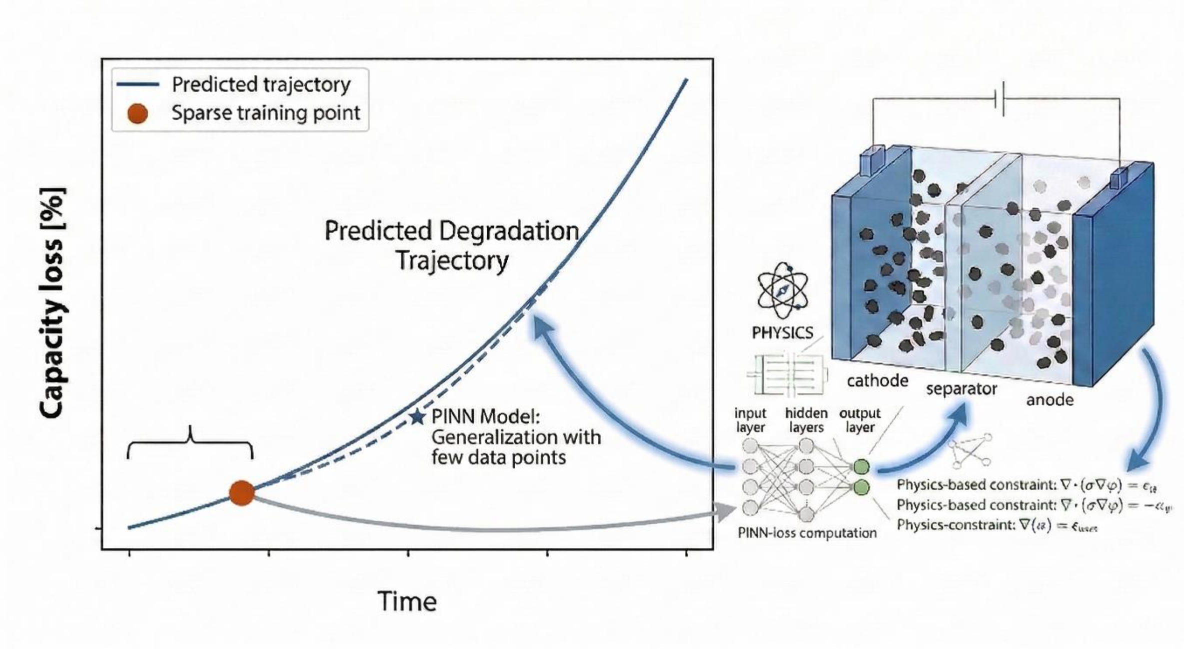

Fig. 1. Physics-informed neural networks enabling the next generation of battery digital twins.

The goal of this project is to compare two approaches:

- **Conventional neural networks (NN)**
  The model learns only from observed data points.
- **Physics-informed neural networks (PINNs)**
  The model learns from both data and physical knowledge encoded in a differential equation.

You will observe that a standard neural network can fit the training points very well, but may fail to extrapolate beyond the observed data. By incorporating physical constraints, the PINN can learn a solution that remains consistent with the underlying physics and therefore produces more reliable predictions outside the training region.

Example Problem

We consider a simplified example representing accelerated battery capacity loss.

The model learns the degradation trajectory over the time domain *t*.

Only a small number of training points are provided. These represent limited experimental measurements of battery capacity loss.

Your task is to investigate how well neural networks and PINNs can reconstruct and extrapolate the aging trajectory.

Workflow Overview

The provided script will guide you through the following steps:

1. Generate synthetic battery aging data
A small number of data points are sampled from the analytical solution.

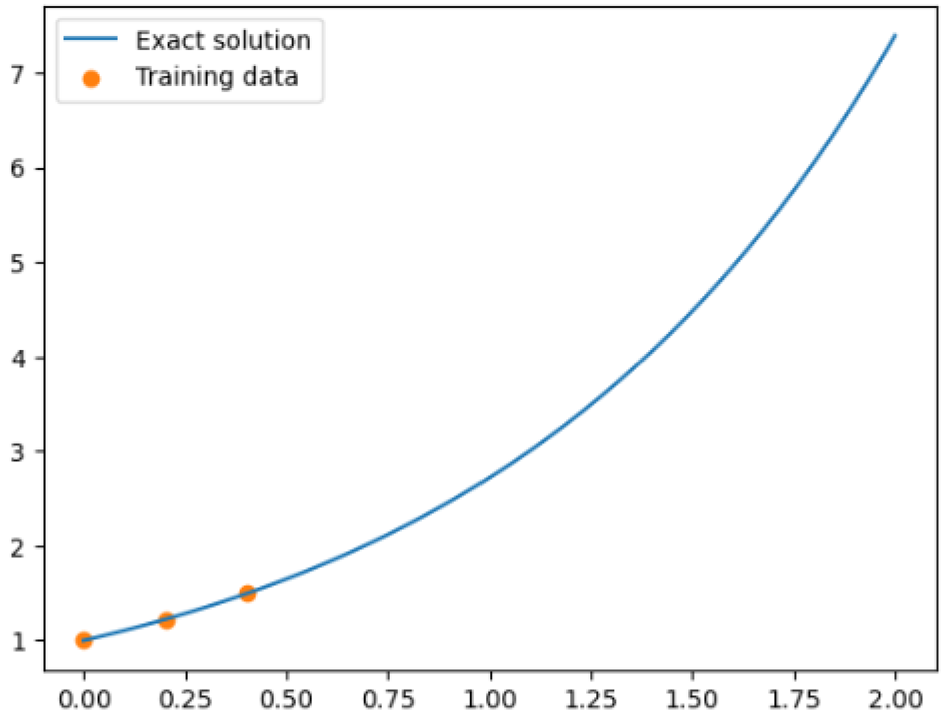

Fig. 1. Generating training data points mimic limited historical data

2. Train a conventional neural network
The neural network learns only from the training data.

3. **Observe the limitation of the conventional neural network**
The network fits the training points but fails to extrapolate the full degradation trajectory.

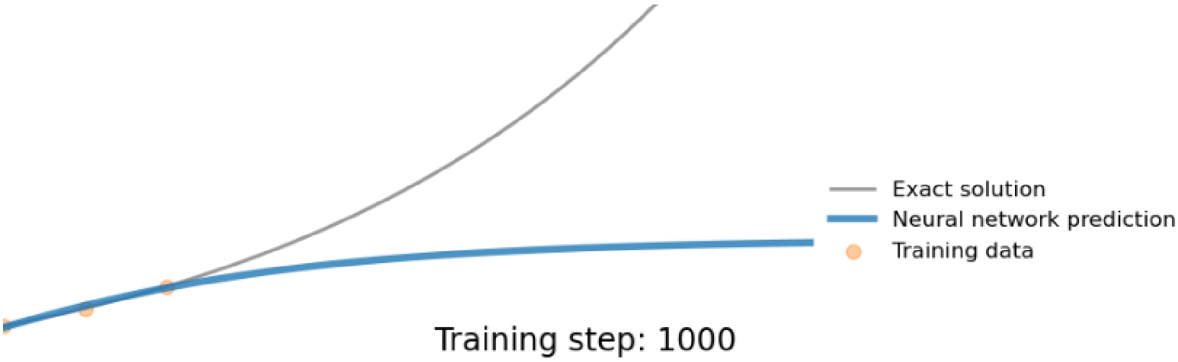

Fig. 2. Conventional neural network prediction results.

4. Train a physics-informed neural network (PINN)
The PINN incorporates the governing differential equation into the loss function.

Fig. 2. Physics-informed neural network prediction results.

5. **Compare the results**
Evaluate how physics constraints improve the prediction.

Tasks

You will use the provided Jupyter Notebook script and explore how different choices affect the model performance.

Run the code and then investigate the following aspects.

1. Neural Network Architecture

Modify the neural network structure:

- number of hidden layers
- number of neurons per layer
- activation functions

Observe how the architecture affects the prediction.

2. Training Iterations

Change the number of training steps.

For example:

- 500
- 5000
- 15000

Does the solution improve with more training?

3. Training Data Size

Increase the number of available training points.

Compare how the neural network and PINN behave when more data is available.

4. Physics vs Data Weight

The PINN loss function combines two components:

$Loss = w_{data}\, L_{data} + w_{physics}\, L_{physics}$

Experiment with different weight values.

Examples:

- physics-dominated training
- data-dominated training
- balanced training

How does this affect the prediction?

5. Prediction Outside the Training Region

Focus on how the models behave **outside the observed data region**.

- Does the neural network extrapolate correctly?
- Does the PINN perform better?

Explain why.

Deliverables

Please briefly summarize:

1. The difference between NN and PINN predictions
2. The effect of changing network architecture
3. The effect of training data size
4. The effect of the physics loss weight

Include plots and short explanations.

Learning Goal

By the end of this project, you should understand:

- how neural networks learn from data
- why extrapolation can fail
- how physical knowledge can improve machine learning models
- the basic idea behind physics-informed neural networks